

LESSON 12.3

MULTIPLYING ALGEBRAIC EXPRESSIONS



LESSON INTRODUCTION

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

MA.8.AR.1.2 Apply Properties of Operations to multiply two linear expressions with rational coefficients.

LEARNING TARGETS

- I can apply properties of operations to multiply two algebraic expressions.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.1 Apply Properties of Operations to add and subtract linear expressions with rational coefficients

WORKING TOWARD

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.

ESSENTIAL QUESTIONS

- How do I apply properties of operations to multiply two algebraic expressions?

LESSON NARRATIVE

This lesson provides students practice with knowledge acquired in prior lessons to apply properties of operations and laws of exponents to multiply two algebraic expressions.

COMPONENT SUMMARY

12.3.1 WARM-UP (~5 MIN.)

Apply the laws of exponents to generate equivalent algebraic expressions

12.3.3 COLLABORATION (10-15 MIN.)

Find and compare equivalent expressions

12.3.5 WRAP-UP (~5 MIN.)

Reflect on understanding of the learning target

12.3.2 GUIDED PRACTICE (10 MIN.)

Multiply monomials by algebraic expressions

12.3.4 ACTIVITY (20-30 MIN.)

Multiply algebraic expressions to find equivalent expressions through a scavenger hunt

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Scavenger Hunt Cards

ADDITIONAL PREPARATION

- Determine student pairing for 12.3.3 Collaboration and 12.3.4 Activity and decide where each group should start in 12.3.4 Activity.

TEACHER TIPS

COMMON MISCONCEPTIONS

- In the Warm-Up, students may divide by 2 to complete the expression instead of multiplying by $\frac{1}{2}$.
- Students may incorrectly try to combine terms. For example, students may try to combine the terms g^3 and $2g$ when given $-\frac{3}{2}g^2(g^3 + 2g)$.
- Students may believe that the different ways an expression can be written are not equivalent. So in 12.2.3 Collaboration, students are writing equivalent expressions, comparing, and discussing expressions with their peers; ensure students don't cross out expressions simply because they aren't written the same way.

THINGS TO CONSIDER

- Teacher-led discussions throughout the unit should include making connections to properties of operations (i.e., the commutative, associative, and distributive properties).

LITERACY AND LANGUAGE ROUTINES

Take Turns

12.3.3 Collaboration and 12.3.4 Activity positions students to work with their peers. Consider using the “Take Turns” Language Routine to encourage students to partake in discussions where they alternate explaining their reasoning and listening to the reasoning of others.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 12.3.3 Collaboration, students will work with a partner to find as many equivalent expressions as they can. Pairs will form groups of 4 to compare their expressions and discuss any differences. This creates an opportunity for students to discuss and justify their thinking with several peers.

DIFFERENTIATE INSTRUCTION

Consider how to use data from Lessons 1 and 2 to determine partners and possible small groups to pull for differentiated support in the 12.3.3 Collaboration and 12.3.4 Activity.

12.3.1 WARM-UP: BELL WORK

TEACHER MOVES

This is a review of MA.8.AR.1.1 from Unit 4.

TEACHER MOVES

Use student understanding to inform your instruction and to determine where to provide more support during this and further lessons. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.



12.3.1 Warm-Up: Bell Work

- Find the product.

$$\frac{8}{3}g^2h^4 \cdot -\frac{5}{2}g^3h$$

$$= -\frac{20}{3}g^5h^5$$

- Fill in the blanks in each expression to make the equation true.

$$-\frac{1}{2}x^3y \cdot 2x^3y^4 = -x^6y^5$$



12.3.2 Guided Practice: Multiplying Monomials by Algebraic Expressions

1. Determine the product of $(-5m^3n^2)$ and $(-2m^2 + \frac{9}{5}n^4)$ using both methods in the table.

Method 1: Use an Area Model	Method 2: Apply the Distributive Property
$-5m^3n^2$ $-2m^2$ $\frac{9}{5}n^4$ $10m^5n^2$ $-9m^3n^6$	$(-5m^3n^2)(-2m^2 + \frac{9}{5}n^4)$ $10m^5n^2 - 9m^3n^6$

2. Work with your partner to find three expressions equivalent to $3x^2(-2x^3 + x)$.
Answers will vary.
 $-6x^5 + 3x^3$
 $3x^3 - 6x^5$
 $(-2x^3 + x) 3x^2$
 $3x^2(x - 2x^3)$

12.3.2 GUIDED PRACTICE: MULTIPLYING MONOMIALS BY ALGEBRAIC EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first two lessons to dictate how “guided” 12.3.2 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



12.3.3 Collaboration: Equivalent Expressions

1. Two expressions are shown in the table. For each expression, work with your partner to find as many equivalent expressions as you can.

Given Expression	$-\frac{3}{2}g^2(g^3 + 2g)$	$4.2q^3r(-6r^3 - 3.01q)$
Equivalent Expressions	$-\frac{3}{2}g^5 - 3g^3$	$-25.2q^3r^4 - 12.642q^4r$
	$-3g^3 - \frac{3}{2}g^5$	$-12.642q^4r - 25.2q^3r^4$
	$(g^3 + 2g) - \frac{3}{2}g^2$	$(-6r^3 - 3.01q) 4.2q^3r$
	$-\frac{3}{2}g^2(2g + g^3)$	$(-3.01q - 6r^3) 4.2q^3r$

2. Find another partner pair and compare your expressions from the previous question. Discuss any differences and add any additional equivalent expressions to your table.

12.3.3 COLLABORATION: EQUIVALENT EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Consider strategic pairing for students who may need additional support. An adaptation to the lesson would be to have students create and compare equivalent expressions one at a time. For students who work at a slower pace, this can help reduce anxiety of not working as fast as others.

TEACHER MOVES

Direct students to add any additional equivalent expressions to their lists, and to cross out any expressions they determined in their discussion that were not equivalent but were on their original list.

12.3.4 ACTIVITY: SCAVENGER HUNT

TEACHER MOVES

Scavenger Hunt

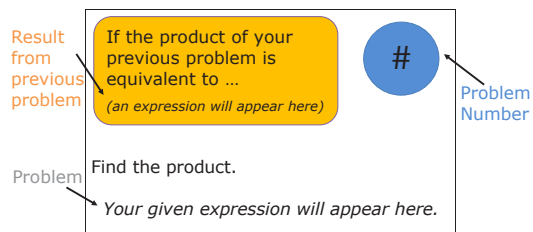
Each group of students should start with a different problem. Tell each group what problem to start on. As groups complete a problem, they will use the product to find the next card to complete. They will complete this process until they have completed all of the scavenger hunt problems and end up at their original problem.

Before starting class, post the scavenger hunt cards around the room, decide on student pairing, and determine which card each pair will start at. Students will frequently have to apply the commutative property of addition to find equivalent expressions.



12.3.4 Activity: Scavenger Hunt

Work with your partner or group to complete the scavenger hunt. Each card in the scavenger hunt appears as shown.



1. Write the number of the problem you start with and the given expression in the first row of the table. Then, work with your group or partner to find the product of the given expression.
2. Look for the card with your product from the previous problem to find your next problem.
3. Repeat this process until your group has completed all the scavenger hunt problems and have ended up back at your original problem.

DIFFERENTIATE INSTRUCTION

This activity provides a kinesthetic entry point as students move throughout the classroom. Utilize the intended flexibility of this lesson to best suit the needs of your students.

Problem Number	Given Expression	Equivalent Expression
1		$-6a^7b^6 + 12a^3b^3$
7		$-2a^6b - \frac{1}{7}a^7b^4$
9		$-14.8a^4b^3$
5		$6ab^3 - 4a^2b$
3		$12a^2b^2 + 1.5a^3b$
6		$-\frac{6}{7}a^3b^2 - \frac{9}{7}ab^3$
2		$12ab^4 + 4a^2b^3$
10		$\frac{2}{7}a^4b^3 + a^5b^3$
8		$-14.8a^6b^2$
4		$\frac{1}{7}ab^3 - 12a^2b^4$

**12.3.5 Wrap-Up: Reflection**

1. Draw an **X** on the line to indicate how you feel about today's learning target.

I can apply properties of operations to multiply two algebraic expressions.

Answers will vary.



I need help. I'm getting there. I understand and have evidence.
I can't multiply two algebraic expressions.

2. Explain your placement of "X" on the scale.

Answers will vary.

12.3.5 WRAP-UP: REFLECTION**QUESTIONING**

What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson and how best to spiral the content.